

标准答案及评分标准

2020年8月20日

一、填空题(每小题 4 分, 共 20 分)

1. $\frac{\sqrt{26}}{3}$

2. $\frac{98}{13}$

3. $\int_0^1 dy \int_0^{\arccos y} f(x, y) dx - \int_{-1}^0 dy \int_{\arccos y}^{\pi} f(x, y) dx$

4. $2R^2$

5. $3 < p \leq 4$

二、计算题(每小题5分, 共20分)

1. 解: 过 $P(1, 2-1)$ 点且垂直于平面 $\pi: 2x - y + z = 5$ 的直线 L 的参数方程为

$$x = 1 + 2t; y = 2 - t; z = -1 + t; \dots\dots\dots (2 \text{ 分})$$

代入平面 π 的方程, 得

$$2 + 4t - 2 + t - 1 + t = 5$$

解得 $t = 1$, 故 P 在平面 π 上投影点的坐标为 $(3, 1, 0)$. $\dots\dots\dots (5 \text{ 分})$

2. 解: $\frac{\partial z}{\partial x} = f'_1 \cdot (1 + \varphi') \qquad \frac{\partial z}{\partial y} = f'_1 \cdot (-\varphi') + f'_2 \qquad \dots\dots\dots (3 \text{ 分})$

$$\frac{\partial^2 z}{\partial x \partial y} = [f''_{11} \cdot (-\varphi') + f''_{12}] (1 + \varphi') - f'_1 \cdot \varphi'' \qquad \dots\dots\dots (5 \text{ 分})$$

3. 解:
$$I = \iiint_V (x + y + z) dx dy dz$$
$$= \int_0^1 dx \int_0^x dy \int_0^{xy} (x + y + z) dz \qquad \dots\dots\dots (3 \text{ 分})$$

$$= \int_0^1 dx \int_0^x [(x + y)xy + \frac{1}{2}x^2y^2] dy$$

$$= \int_0^1 [\frac{1}{2}x^4 + \frac{1}{3}x^4 + \frac{1}{6}x^5] dx = \frac{7}{36}. \qquad \dots\dots\dots (5 \text{ 分})$$

4. 解: $\text{gradu} = (\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}) = (2x, z, y)$ (2分)

$$\text{div}(\text{gradu}) = \text{div}(2x, z, y)$$

$$= \frac{\partial}{\partial x}(2x) + \frac{\partial}{\partial y}(z) + \frac{\partial}{\partial z}(y)$$

$$= 2. \quad \text{.....(5 分)}$$

三、解: 记 $\int_0^t u f(u^2 + t^2) du = g(t), \quad F(x) = \int_0^x g(t) dt,$

故 $F'(x) = g(x) = \int_0^x u f(u^2 + x^2) du$ (3 分)

令 $y = u^2 + x^2$, 则 $F'(x) = \frac{1}{2} \int_{x^2}^{2x^2} f(y) dy,$ (6 分)

得 $F''(x) = 2xf(2x^2) - xf(x^2).$ (8 分)

四、解: 所求转动惯量 $I = \iiint_{\Omega} (x^2 + y^2) dx dy dz$ (1 分)

做球面坐标变换 $\begin{cases} x = \rho \sin \varphi \cos \theta \\ y = \rho \sin \varphi \sin \theta \\ z = \rho \cos \theta \end{cases} \quad \frac{\partial(x, y, z)}{\partial(\rho, \varphi, \theta)} = \rho^2 \sin \varphi$

Ω 边界曲面的球坐标方程分别为 $\varphi = \frac{\pi}{4}$ 和 $\rho = 2 \cos \varphi$

$$I = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{2\cos\varphi} \rho^2 \sin^2 \varphi \rho^2 \sin \varphi d\rho \quad \text{..... (4 分)}$$

$$= 2\pi \cdot \frac{32}{5} \int_0^{\frac{\pi}{4}} \cos^5 \varphi \sin^3 \varphi d\varphi$$

$$= \frac{64\pi}{5} \int_0^{\frac{\pi}{4}} \cos^5 \varphi (\cos^2 \varphi - 1) d \cos \varphi$$

$$= \frac{64\pi}{5} \left(\frac{u^8}{8} - \frac{u^6}{6} \right) \Big|_1^{\frac{\sqrt{2}}{2}} = \frac{11\pi}{30} \quad \text{..... (6 分)}$$

(注: 柱坐标计算同样可得, 评分标准参考以上球坐标的方法)

五、解: $f'_x = ay^2 + 3cx^2z^2 \quad f'_y = 2axy + bz \quad f'_z = by + 2cx^3z$

$$\text{grad} f(M) = \{4a + 3c, 4a - b, 2b - 2c\} \quad \text{.....(3 分)}$$

$$4a + 3c = 0 \quad 4a - b = 0 \quad 2b - 2c = 64 \quad \text{.....(6 分)}$$

解得 $a = 6 \quad b = 24 \quad c = -8$ (8 分)

六、解：1). $P(x, y) = (x - y)(x^2 + y^2)^\lambda, Q(x, y) = (x + y)(x^2 + y^2)^\lambda$ (1 分)

$$\frac{\partial P}{\partial y} = -(x^2 + y^2)^\lambda + 2y\lambda(x - y)(x^2 + y^2)^{\lambda-1}$$

$$\frac{\partial Q}{\partial x} = (x^2 + y^2)^\lambda + 2x\lambda(x + y)(x^2 + y^2)^{\lambda-1} \quad \text{.....(3 分)}$$

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow 2(\lambda + 1)(x^2 + y^2)^\lambda = 0 \Rightarrow \lambda = -1 \quad \text{.....(4 分)}$$

$$2). P(x, y) = \frac{x - y}{x^2 + y^2}, Q(x, y) = \frac{x + y}{x^2 + y^2}$$

$$df(x, y) = P(x, y)dx + Q(x, y)dy \quad \text{.....(5 分)}$$

$$f(1, \sqrt{3}) - f(2, 0) = \int_{(2,0)}^{(1,\sqrt{3})} P(x, y)dx + Q(x, y)dy \quad \text{.....(6 分)}$$

$$= \int_0^{\sqrt{3}} Q(1, y)dy + \int_2^1 P(x, 0)dx = \int_0^{\sqrt{3}} \frac{1+y}{1+y^2}dy + \int_2^1 \frac{x}{x^2}dx$$

$$= \arctan y \Big|_0^{\sqrt{3}} + \frac{1}{2} \ln(1+y^2) \Big|_0^{\sqrt{3}} - \ln 2$$

$$= \frac{\pi}{3} \quad \text{.....(8 分)}$$

七、解： $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}(x)}{u_n(x)} \right| = \lim_{n \rightarrow \infty} \frac{n(2n-1)}{(n+1)(2n+1)} x^2 = x^2$

当 $x^2 < 1$, 即 $|x| < 1$ 时级数收敛.

故收敛区间为: $-1 < x < 1$ (2 分)

设
$$S(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n(2n-1)} x^{2n}$$

$$S'(x) = \sum_{n=1}^{\infty} \frac{2(-1)^{n-1}}{(2n-1)} x^{2n-1}$$

$$S''(x) = \sum_{n=1}^{\infty} 2(-1)^{n-1} x^{2n-2} \quad \text{..... (4 分)}$$

$$= \sum_{n=1}^{\infty} 2(-x^2)^{n-1} = \frac{2}{1+x^2} \quad \text{..... (6 分)}$$

$$S'(x) = 2 \arctan x$$

$$S(x) = 2x \arctan x - \ln(1+x^2) \quad \dots\dots\dots (8 \text{ 分})$$

八、解： $f(x) = \frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2} \quad \dots\dots\dots (2 \text{ 分})$

$$= \frac{1}{x-1+2} - \frac{1}{x-1+3} = \frac{1}{2} \cdot \frac{1}{1+\frac{x-1}{2}} - \frac{1}{3} \cdot \frac{1}{1+\frac{x-1}{3}}$$

$$= \frac{1}{2} \sum_{n=1}^{\infty} (-1)^{n-1} \left(\frac{x-1}{2}\right)^{n-1} - \frac{1}{3} \sum_{n=1}^{\infty} (-1)^{n-1} \left(\frac{x-1}{3}\right)^{n-1}$$

$$= \sum_{n=1}^{\infty} (-1)^{n-1} \left(\frac{1}{2^n} - \frac{1}{3^n}\right) (x-1)^{n-1} \quad \dots\dots\dots (5 \text{ 分})$$

收敛区间 $(-1, 3)$ $\dots\dots\dots (6 \text{ 分})$

$$f^{(5)}(1) = -5! \left(\frac{1}{2^6} - \frac{1}{3^6}\right) \quad \dots\dots\dots (8 \text{ 分})$$

九、解： 记 $S_1: z = R, x^2 + y^2 \leq R^2$, 取上侧； $S_2: z = -R, x^2 + y^2 \leq R^2$, 取下侧；

S_3 : S 的侧面 (即圆柱面部分) ；

$S_{3\text{前}}: x = \sqrt{R^2 - y^2}, -R \leq y \leq R, -R \leq z \leq R$, 取前侧；

$S_{3\text{后}}: x = -\sqrt{R^2 - y^2}, -R \leq y \leq R, -R \leq z \leq R$, 取后侧；

$$\begin{aligned} \iint_S \frac{xdydz}{x^2 + y^2 + z^2} &= \iint_{S_1} \frac{xdydz}{x^2 + y^2 + z^2} + \iint_{S_2} \frac{xdydz}{x^2 + y^2 + z^2} + \iint_{S_3} \frac{xdydz}{x^2 + y^2 + z^2} \\ &= \iint_{S_{3\text{前}}} \frac{xdydz}{x^2 + y^2 + z^2} + \iint_{S_{3\text{后}}} \frac{xdydz}{x^2 + y^2 + z^2} \quad \dots\dots\dots (2 \text{ 分}) \end{aligned}$$

$$= \iint_{D_{yz}} \frac{\sqrt{R^2 - y^2} dydz}{R^2 + z^2} - \iint_{D_{yz}} \frac{-\sqrt{R^2 - y^2} dydz}{R^2 + z^2}$$

$$= 2 \int_{-R}^R \sqrt{R^2 - y^2} dy \int_{-R}^R \frac{dz}{R^2 + z^2} = \frac{\pi^2 R}{2}$$

其中 $D_{yz} = \{(y, z) | -R \leq y \leq R, -R \leq z \leq R\}$. $\dots\dots\dots (5 \text{ 分})$

$$\iint_S \frac{z^2}{x^2 + y^2 + z^2} dxdy = \iint_{S_1} \frac{z^2}{x^2 + y^2 + z^2} dxdy + \iint_{S_2} \frac{z^2}{x^2 + y^2 + z^2} dxdy + \iint_{S_3} \frac{z^2}{x^2 + y^2 + z^2} dxdy$$

$$\begin{aligned}
&= \iint_{S_1} \frac{z^2}{x^2 + y^2 + z^2} dx dy + \iint_{S_2} \frac{z^2}{x^2 + y^2 + z^2} dx dy \\
&= \iint_{D_{xy}} \frac{R^2}{x^2 + y^2 + R^2} dx dy - \iint_{D_{xy}} \frac{(-R)^2}{x^2 + y^2 + R^2} dx dy \\
&= 0 \quad \dots\dots\dots (7 \text{ 分})
\end{aligned}$$

所以, 原式 $I = \frac{\pi^2 R}{2}$. \dots\dots\dots (8 \text{ 分})

十、解: 1). $F(t) = \int_0^{2\pi} d\theta \int_0^\pi d\phi \int_0^t \rho^2 f(\rho^2) \sin \phi d\rho$ \dots\dots\dots (1 \text{ 分})

$$= 4\pi \int_0^t \rho^2 f(\rho^2) d\rho$$

$$F'(t) = 4\pi t^2 f(t^2) \quad \dots\dots\dots (3 \text{ 分})$$

2). $\sum_{n=1}^{\infty} n^{1-\lambda} F'(\frac{1}{n}) = \sum_{n=1}^{\infty} 4\pi \frac{1}{n^{1+\lambda}} f(\frac{1}{n^2})$ \dots\dots\dots (4 \text{ 分})

因 $f(0) \neq 0$ 且 $f(x)$ 在 0 点连续, $f(x)$ 在 0 点右小邻域有局部保号性,
此时 $f(x)$ 要么正, 要么负.

$$\lim_{n \rightarrow \infty} \frac{\frac{4\pi}{n^{1+\lambda}} f(\frac{1}{n^2})}{\frac{1}{n^{1+\lambda}}} = 4\pi f(0)$$

$\lambda > 0$ 时收敛, $\lambda \leq 0$ 时发散。 \dots\dots\dots (6 \text{ 分})